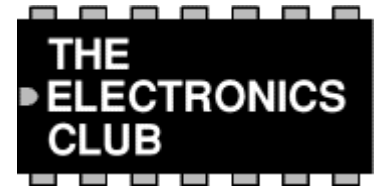


[Home](#) | [Capacitor](#) | [Connector](#) | [Diode](#) | [IC](#) | [Lamp](#) | [LED](#) | [Relay](#) | [Resistor](#) | [Switch](#) | [Transistor](#) | [Variable Resistor](#) | [Other](#)

Heat sinks for transistors



Heat sinks are needed for [transistors](#) passing large currents.

Why is a heat sink needed?

Waste heat is produced in transistors due to the current flowing through them. If you find that a transistor is becoming too hot to touch it certainly needs a heat sink! The heat sink helps to dissipate (remove) the heat by transferring it to the surrounding air.



Heat sink

Photograph © [Rapid Electronics](#)

The rate of producing waste heat is called the thermal power, P . Usually the base current I_B is too small to contribute much heat, so the thermal power is determined by the collector current I_C and the voltage V_{CE} across the transistor:

$$P = I_C \times V_{CE} \quad (\text{see diagram below})$$

The heat is not a problem if I_C is small or if the transistor is used as a switch because when 'full on' V_{CE} is almost zero. However, power transistors used in circuits such as an audio amplifier or a motor speed controller will be partly on most of the time and V_{CE} may be about half the supply voltage. These power transistors will almost certainly need a heat sink to prevent them overheating.



Insulation kit



Heat-conducting paste

Photographs © [Rapid Electronics](#)

Power transistors usually have bolt holes for attaching heat sinks, but clip-on heat sinks are also available. Make sure you use the right type for your transistor. Many transistors have metal cases which are connected to one of their leads so it may be necessary to insulate the heat sink from the transistor. Insulating kits are available with a mica sheet and a plastic sleeve for the bolt. Heat-conducting paste can be used to improve heat flow from the transistor to the heat sink, this is especially important if an insulation kit is used.

Heat sink ratings

Heat sinks are rated by their [thermal resistance](#) (R_{th}) in $^{\circ}\text{C}/\text{W}$. For example $2^{\circ}\text{C}/\text{W}$ means the heat sink (and therefore the component attached to it) will be 2°C hotter than the surrounding air for every 1W of heat it is dissipating. Note that **a lower thermal resistance means a better heat sink.**

This is how you work out the required heat sink rating:



1. Work out thermal power to be dissipated, $P = I_C \times V_{CE}$
If in doubt use the largest likely value for I_C and assume that V_{CE} is half the supply voltage.
For example if a power transistor is passing 1A and connected to a 12V supply, the power P is about $1 \times \frac{1}{2} \times 12 = 6W$.
2. Find the maximum operating temperature (T_{max}) for the transistor if you can, otherwise assume $T_{max} = 100^\circ C$.
3. Estimate the maximum ambient (surrounding air) temperature (T_{air}). If the heat sink is going to be outside the case $T_{air} = 25^\circ C$ is reasonable, but inside it will be higher (perhaps $40^\circ C$) allowing for everything to warm up in operation.
4. Work out the maximum thermal resistance (R_{th}) for the heat sink using: $R_{th} = (T_{max} - T_{air}) / P$
With the example values given above: $R_{th} = (100-25)/6 = 12.5^\circ C/W$.
5. Choose a heat sink with a thermal resistance which is **less** than the value calculated above (remember lower value means better heat sinking!) for example $5^\circ C/W$ would be a sensible choice to allow a safety margin. A $5^\circ C/W$ heat sink dissipating 6W will have a temperature difference of $5 \times 6 = 30^\circ C$ so the transistor temperature will rise to $25 + 30 = 55^\circ C$ (safely less than the $100^\circ C$ maximum).
6. All the above assumes the transistor is at the same temperature as the heat sink. This is a reasonable assumption if they are firmly bolted or clipped together. However, you may have to put a mica sheet or similar between them to provide electrical insulation, then the transistor will be hotter than the heat sink and the calculation becomes more difficult. For typical mica sheets you should subtract $2^\circ C/W$ from the thermal resistance (R_{th}) value calculated in step 4 above.

If this all seems too complex you can try attaching a moderately large heat sink and hope for the best. Cautiously monitor the transistor temperature with your finger, if it becomes painfully hot switch off immediately and use a larger heat sink!

Why thermal resistance?

The term '**thermal resistance**' is used because it is analagous to electrical resistance:

- The temperature difference across the heat sink (between the transistor and air) is like voltage (potential difference) across a resistor.
- The thermal power (rate of heat) flowing through the heat sink from transistor to air is like current flowing through a resistor.
- So $R = V/I$ becomes $R_{th} = (T_{max} - T_{air})/P$
- Just as you need a voltage difference to make current flow, you need a temperature difference to make heat flow.

[Home](#) | [Capacitor](#) | [Connector](#) | [Diode](#) | [IC](#) | [Lamp](#) | [LED](#) | [Relay](#) | [Resistor](#) | [Switch](#) | [Transistor](#) | [Variable Resistor](#) | [Other](#)

- [Electronics Club Home Page](#)
- [Site Map](#)
- [Example Projects](#)
- [Construction of Projects](#)
- [Soldering Guide](#)
- [Study Electronics](#)
- [Electronic Components](#)



- [555 Timer](#)
- [Circuit Symbols](#)
- [Frequently Asked Questions](#)
- [Links to other Electronics sites](#)

© John Hewes 2007, The Electronics Club, www.kpsec.freeuk.com